

Zadanie

Pierwszy capture w Wireshark — przechwycenie ruchu z własnego komputera i identyfikacja warstw OSI w realnych pakietach.

```
# adres MAC na macOS
ifconfig en0 | grep ether
```

PlikEdycjaWidokPrzejdźPrzechwytywanieAnalizaStatystykiTelefoniaBezprzewodoweNarzędziaPomoc

dns

No.	Time	Source	Destination	Protocol	Length	Info
28	6.329623200	192.168.1.226	192.168.1.1	DNS	91	Standard query 0x3d0d A signaler-pa.clients6.google.com
29	6.363652700	192.168.1.226	192.168.1.1	DNS	91	Standard query 0x3d0d A signaler-pa.clients6.google.com
31	6.374879500	192.168.1.1	192.168.1.226	DNS	107	Standard query response 0x3d0d A signaler-pa.clients6.google.com
32	6.375780500	192.168.1.226	192.168.1.1	DNS	91	Standard query 0x3284 AAAA signaler-pa.clients6.google.com
33	6.410321500	192.168.1.226	192.168.1.1	DNS	91	Standard query 0x3284 AAAA signaler-pa.clients6.google.com
35	6.413429900	192.168.1.1	192.168.1.226	DNS	119	Standard query response 0x3284 AAAA signaler-pa.clients6.google.com
65	12.095363700	192.168.1.226	192.168.1.1	DNS	82	Standard query 0x679e HTTPS configcat.supabase.com
66	12.095620800	192.168.1.226	192.168.1.1	DNS	82	Standard query 0x0902 A configcat.supabase.com
68	12.127056300	192.168.1.1	192.168.1.226	DNS	152	Standard query response 0x679e HTTPS configcat.supabase.com
70	12.131387600	192.168.1.1	192.168.1.226	DNS	114	Standard query response 0x0902 A configcat.supabase.com
71	12.132962200	192.168.1.226	192.168.1.1	DNS	82	Standard query 0xc512 AAAA configcat.supabase.com
76	12.162236100	192.168.1.1	192.168.1.226	DNS	138	Standard query response 0xc512 AAAA configcat.supabase.com
220	26.843107000	192.168.1.226	192.168.1.1	DNS	76	Standard query 0x7ae2 A www.linkedin.com

> Frame 28: Packet, 91 bytes on wire (728 bits), 91 bytes captured (728 bits) on interface 0 (Device\NPF_{E01C0B12-47A6-40CB-991B-E276CAAD2})

> Ethernet II, Src: LCFCElectron_0d:98:5d (84:a9:38:0d:98:5d), Dst: zt...

> Internet Protocol Version 4, Src: 192.168.1.226, Dst: 192.168.1.1

> User Datagram Protocol, Src Port: 49363, Dst Port: 53

> Domain Name System (query)

0000d4 72 26 8d 8f a0 84 a9 38 0d 98 5d 08 00 45 00r&....

001000 4d 54 0a 00 00 80 11 00 00 c0 a8 01 e2 c0 a8MT....

002001 01 c0 d3 00 35 00 39 84 7e 3d 0d 01 00 00 01....5.

003000 00 00 00 00 00 0b 73 69 67 6e 61 6c 65 72 2d.....

004070 61 08 63 6c 69 65 6e 74 73 36 06 67 6f 6f 67pa-clie

00506c 65 03 63 6f 6d 00 00 01 00 01le.com

```
Frame 28: Packet, 91 bytes on wire (728 bits), 91 bytes captured (
  Section number: 1
> Interface id: 0 (\Device\NPF_{E01C0B12-47A6-40CB-991B-E276CAAD2}
  Encapsulation type: Ethernet (1)
  Arrival Time: Aug 12, 2026 16:06:47.298703000 Środkowoeuropejsk
  UTC Arrival Time: Aug 12, 2026 14:06:47.298703000 UTC
  Epoch Arrival Time: 1786543607.298703000
  [Time shift for this packet: 0.000000000 seconds]
  [Time delta from previous captured frame: 10.662000 millisecond
  [Time since reference or first frame: 6.329623200 seconds]
  Frame Number: 28
  Frame Length: 91 bytes (728 bits)
  Capture Length: 91 bytes (728 bits)
  [Frame is marked: False]
```

```
▼ Internet Protocol Version 4, Src: 192.168.1.226, Dst: 192.168.1.1
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 77
    Identification: 0x540a (21514)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0x0000 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 192.168.1.226
    Destination Address: 192.168.1.1
```

Domain Name System (query)

```
Transaction ID: 0x3d0d
> Flags: 0x0100 Standard query
Questions: 1
Answer RRs: 0
Authority RRs: 0
Additional RRs: 0
> Queries
\[Response In: 31\]
```

Analiza — pakiet DNS query

L2 – Ethernet II (Data Link)

- Source: LCFCElectron_0d:98:5d (84:a9:38:0d:98:5d)
- Destination: zte_8d:8f:a0 (d4:72:26:8d:8f:a0)
- Type: IPv4 (0x0800)

L3 – Internet Protocol (Network)

- Source: 192.168.1.226
- Destination: 192.168.1.1
- TTL: 128
- Protocol: UDP (17)

L4 – UDP (Transport)

- Src Port: 49363
- Dst Port: 53

L7 – DNS (Application)

- Transaction ID: 0x3d0d
- Standard query, 1 pytanie
- [\[Response In: 31\]](#)

Encapsulation:

Ethernet [IP [UDP [DNS]]]

Windows IP Configuration

Host Name : Lenovo
Primary Dns Suffix :
Node Type : Hybrid
IP Routing Enabled. : No
WINS Proxy Enabled. : No

Ethernet adapter Ethernet:

Connection-specific DNS Suffix . : lan
Description : Intel(R) Ethernet Connection (10) I219-V
Physical Address. : 84-A9-38-0D-98-5D
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::ae75:e628:30cc:7656%16(Preferred)
IPv4 Address. : 192.168.1.226(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : poniedziałek, 10 sierpnia 2026 13:24:24
Lease Expires : czwartek, 13 sierpnia 2026 02:28:52
Default Gateway : 192.168.1.1
DHCP Server : 192.168.1.1
DHCPv6 IAID : 92580152
DHCPv6 Client DUID. : 00-01-01-00-31-5E-B3-82-84-A9-38-0D-98-5D
DNS Servers : 192.168.1.1
NetBIOS over Tcpi. : Enabled

Ethernet adapter Ethernet 2:

Connection-specific DNS Suffix . :
Description : VirtualBox Host-Only Ethernet Adapter
Physical Address. : 0A-00-27-00-00-35
DHCP Enabled. : No
Autoconfiguration Enabled : Yes
Link-local IPv6 Address : fe80::6f64:2494:4b5e:cf12%53(Preferred)
IPv4 Address. : 192.168.56.1(Preferred)
Subnet Mask : 255.255.255.0
Default Gateway :
DHCPv6 IAID : 889847847
DHCPv6 Client DUID. : 00-01-01-00-31-5E-B3-82-84-A9-38-0D-98-5D
NetBIOS over Tcpi. : Enabled